2025.12.16. 多元函数微分 II.

1. 梯度和方向导数计算.

结论: 对 $|l|=1$, $\frac{\partial f}{\partial l} = \nabla f^T l$

① 沿 ∇f, f 上升最快

② 沿 ∇f 反向, f 下降最快

③ 垂直于 ∇f, f 不变 (梯度 $\perp$ 等高线)

④ $\frac{\partial f}{\partial l} \in [-|\nabla f|, |\nabla f|]$.

① $F(x,y,z) = z^3 + x^2 z - 2y^3 = 0$. 求 $z(x,y)$ 在 $(1,1)$ 处最大方向导数.

$$3z^2 dz + 2xz\,dx + x^2 dz - 6y^2 dy = 0.$$

$$\Rightarrow dz = \frac{-2xz\,dx + 6y^2 dy}{3z^2 + x^2} = -\frac{1}{2}dx + \frac{3}{2}dy$$

故 $\nabla z = \begin{pmatrix} -\frac{1}{2} \\ \frac{3}{2} \end{pmatrix}$. 知最大方向导数为 $|\nabla z| = \frac{\sqrt{10}}{2}$.

② $f(x,y,z) = \left(\frac{2x}{z}\right)^y$ $(z \neq 0)$. 求 $(\frac{1}{2},1,1)$ 处最速下降方向.

$$\nabla f = \begin{pmatrix} y\left(\frac{2x}{z}\right)^{y-1} \cdot \frac{2}{z} \\ \left(\frac{2x}{z}\right)^y \ln\frac{2x}{z} \\ (2x)^y \cdot (-y) \cdot z^{-(y+1)} \end{pmatrix}$$

在 $(\frac{1}{2},1,1)$ 处为 $\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$

$\Rightarrow$ 最速下降为 $-\nabla f$ 反向, 即 $\begin{pmatrix} -2\sqrt{5}/5 \\ 0 \\ \sqrt{5}/5 \end{pmatrix}$.

2. 验证 PDE.

① 任何函数. $u = F(x^2+y^2)$ 满足 $(y\partial_x - x\partial_y)u = 0$.

证明: $\partial_x u = 2xF'$, $\partial_y u = 2yF'$

$\Rightarrow LHS = 2xyF' - 2xyF' = 0$ □.

② 扩散方程/热方程. $f(x,t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}$ 满足 $\partial_t f = \frac{1}{2}\partial_x^2 f$.

证明: $\partial_t f = \frac{1}{\sqrt{2\pi}}\left(-\frac{1}{2t^{3/2}}\right)e^{-\frac{x^2}{2t}} + \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}\left(\frac{x^2}{2t^2}\right) = -\frac{1}{2t}f + \frac{x^2}{2t^2}f$

$\partial_x f = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}\left(-\frac{x}{t}\right) = -\frac{x}{t}f$

$\partial_x^2 f = -\frac{1}{t}f - \frac{x}{t}\partial_x f = -\frac{1}{t}f + \frac{x^2}{t^2}f = 2\partial_t f$. □.

③ 输运方程. $f(x,t)=u(x-ct)$ 满足 $\partial_t f + c\partial_x f = 0$

证明: $\partial_t f = -cu'$. $\partial_x u = u'$

$\Rightarrow \partial_t f + c\partial_x f = -cu' + cu' = 0$ □.

④ 生灭方程. $f(t) = e^{rt}$ 满足 $\partial_t f = rf$

证明: $\partial_t f = re^{rt} = rf$ □.

⑤ Fokker-Plank 方程(带源项)

$f(x,t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{(x-ct)^2}{2t}+rt}$ 满足 $\partial_t f + c\partial_x f = \frac{1}{2}\partial_x^2 f + rf$.

证明: 令 $g(y,t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{y^2}{2t}}$ 则 $f(x,t) = e^{rt}\cdot g(x-ct)$

则 $\partial_t f = re^{rt}\cdot g(x-ct) + e^{rt}\cdot g_2(x-ct,t) + e^{rt}\cdot g_1(x-ct,t)\cdot(-c)$

由②知 $g_2 = \frac{1}{2} g_{11}$

知 $\partial_t f = rf + \frac{1}{2}e^{rt} g_{11} - ce^{rt} g_1$. 又 $g_1 e^{rt} = f_1$ (易验证).

$g_{11} e^{rt} = f_{11}$.

故 $\partial_t f = rf + \frac{1}{2} f_{11} - cf_1$ 即 $\partial_t f + c\partial_x f = \frac{1}{2}\partial_x^2 f + rf$. □.

3. 微分算符的复杂运算.

① $f, g \in C^2$ (连续二阶偏导数). $h(x,y) \triangleq xg(\frac{y}{x}) + f(\frac{y}{x})$.

求 $x^2 h_{xx} + 2xy h_{xy} + y^2 h_{yy}$.

解: $h_x = g - \frac{y}{x} g' - \frac{y}{x^2} f'$, $h_y = g' + \frac{1}{x} f'$

$h_{xx} = \frac{y^2}{x^3} g'' + \frac{2y}{x^3} f' + \frac{y^2}{x^4} f''$

$h_{xy} = -\frac{y}{x^2} g'' - \frac{1}{x^2} f' - \frac{y}{x^3} f''$

$h_{yy} = \frac{1}{x} g'' + \frac{1}{x^2} f''$.

代入即证□.

另一种理解：做极换元 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$，则 $h(x,y) = r\cdot\cos\theta\, g(\tan\theta) + f(\tan\theta)$

形如 $A(\theta)\cdot r + B(\theta)$

$$\partial_r = \frac{\partial x}{\partial r}\partial_x + \frac{\partial y}{\partial r}\partial_y$$

$$= \cos\theta\,\partial_x + \sin\theta\,\partial_y$$

$$\partial_r^2 = \partial_r\partial_r = \partial_r(\cos\theta\,\partial_x + \sin\theta\,\partial_y)$$

$$= \cos\theta\,\partial_r\partial_x + \sin\theta\,\partial_r\partial_y$$

（θ 与 r 无关）

$$= \cos\theta(\cos\theta\,\partial_x + \sin\theta\,\partial_y)\partial_x + \sin\theta(\cos\theta\,\partial_x + \sin\theta\,\partial_y)\partial_y$$

$$= \cos^2\theta\,\partial_x^2 + 2\sin\theta\cos\theta\,\partial_x\partial_y + \sin^2\theta\,\partial_y^2$$

混合二阶偏导可交换顺序. $\partial_x\partial_y = \partial_y\partial_x$.

$$\Rightarrow r^2\partial_r^2 = x^2\partial_x^2 + 2xy\,\partial_x\partial_y + y^2\partial_y^2$$

故 $x^2 h_{xx} + 2xy\,h_{xy} + y^2 h_{yy} = r^2 h_{rr}$

$$= r^2\cdot\partial_r^2(A(\theta)+r+B(\theta))$$

$$= r^2\cdot 0 = 0. \qquad \square$$

② $\begin{cases} \partial_r = \cos\theta\,\partial_x + \sin\theta\,\partial_y \\ \partial_\theta = y\,\partial_x - x\,\partial_y \end{cases} \rightsquigarrow r\partial_r = x\partial_x + y\partial_y.$

(i) $xf_x + yf_y = 0 \rightsquigarrow f(x,y) = F(\theta)$ （切向函数）

因为 $\partial_r f(x,y) = (xf_x + yf_y)/r = 0$.

Rmk. 考虑 $(0,0)$ 处连续性，这种函数只能是常数

因为 $\forall\theta\in\mathbb{R}$. 有 $F(\theta) = f(0,0)$

(ii) $yf_x - xf_y = 0 \rightsquigarrow f(x,y) = G(r)$ （径向函数）

因为 $\partial_\theta f(x,y) = yf_x - xf_y = 0$. $\square$.

③ Laplace 算子的极坐标形式.

$$\partial_x^2 u + \partial_y^2 u = \partial_r^2 u + \frac{1}{r}\partial_r u + \frac{1}{r^2}\partial_\theta^2 u.$$

证明: $\begin{cases} \partial_x = \frac{\partial r}{\partial x}\partial_r + \frac{\partial \theta}{\partial x}\partial_\theta \\ \partial_y = \frac{\partial r}{\partial y}\partial_r + \frac{\partial \theta}{\partial y}\partial_\theta . \end{cases}$

$r = \sqrt{x^2+y^2}$, $\theta = \arctan\frac{y}{x}$.

$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \cos\theta$, $\frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \sin\theta$

$\frac{\partial \theta}{\partial x} = -\frac{y}{x^2+y^2} = -\frac{\sin\theta}{r}$, $\frac{\partial \theta}{\partial y} = \frac{x}{x^2+y^2} = \frac{\cos\theta}{r}$

$$\Rightarrow \begin{cases} \partial_x = \cos\theta\,\partial_r - \frac{\sin\theta}{r}\partial_\theta \\ \partial_y = \sin\theta\,\partial_r + \frac{\cos\theta}{r}\partial_\theta . \end{cases}$$

(i) $\partial_x^2 = \partial_x\partial_x = \cos\theta\,\partial_r\left(\cos\theta\,\partial_r - \frac{\sin\theta}{r}\partial_\theta\right) - \frac{\sin\theta}{r}\partial_\theta\left(\cos\theta\,\partial_r - \frac{\sin\theta}{r}\partial_\theta\right)$

$= \cos^2\theta\,\partial_r^2 - \cos\theta\sin\theta\cdot\partial_r\left(\frac{1}{r}\right)\partial_\theta - \frac{\sin\theta\cos\theta}{r}\partial_r\partial_\theta$

$- \frac{\sin\theta}{r}(\partial_\theta\cos\theta)\partial_r - \frac{\sin\theta\cos\theta}{r}\partial_\theta\partial_r - \frac{\sin^2\theta}{r^2}(\partial_\theta\sin\theta)\partial_\theta + \frac{\sin^2\theta}{r^2}\partial_\theta^2$

$= \cos^2\theta\,\partial_r^2 + \frac{\sin\theta\cos\theta}{r^2}\partial_\theta - \frac{\sin\theta\cos\theta}{r}\partial_r\partial_\theta + \frac{\sin^2\theta}{r}\partial_r - \frac{\sin\theta\cos\theta}{r}\partial_r\partial_\theta$

$+ \frac{\sin\theta\cos\theta}{r^2}\partial_\theta + \frac{\sin^2\theta}{r^2}\partial_\theta^2$

$= \cos^2\theta\,\partial_r^2 + \frac{\sin^2\theta}{r^2}\partial_\theta^2 - \frac{2\sin\theta\cos\theta}{r}\partial_r\partial_\theta + \frac{2\sin\theta\cos\theta}{r^2}\partial_\theta + \frac{\sin^2\theta}{r}\partial_r$

(ii) $\partial_y^2 = \partial_y\partial_y = \sin\theta\,\partial_r\left(\sin\theta\,\partial_r + \frac{\cos\theta}{r}\partial_\theta\right) + \frac{\cos\theta}{r}\partial_\theta\left(\sin\theta\,\partial_r + \frac{\cos\theta}{r}\partial_\theta\right)$

$= \sin^2\theta\,\partial_r^2 - \frac{\sin\theta\cos\theta}{r^2}\partial_\theta + \frac{\sin\theta\cos\theta}{r}\partial_r\partial_\theta + \frac{\cos^2\theta}{r}\partial_r + \frac{\sin\theta\cos\theta}{r}\partial_r\partial_\theta$

$- \frac{\cos\theta\sin\theta}{r^2}\partial_\theta + \frac{\cos^2\theta}{r^2}\partial_\theta^2$

$= \sin^2\theta\,\partial_r^2 + \frac{\cos^2\theta}{r^2}\partial_\theta^2 + \frac{2\sin\theta\cos\theta}{r}\partial_r\partial_\theta - \frac{2\sin\theta\cos\theta}{r^2}\partial_\theta + \frac{\cos^2\theta}{r}\partial_r$

$\Rightarrow \partial_x^2 + \partial_y^2 = \partial_r^2 + \frac{1}{r^2}\partial_\theta^2 + \frac{1}{r}\partial_r$. $\square$

④. $x = e^s\cos t,\quad y = e^s\sin t.$

$\partial_x^2 u + \partial_y^2 u = 0.$ 求证: $\partial_s^2 u + \partial_t^2 u = 0.$

证明: $\partial_s = \frac{\partial x}{\partial s}\partial_x + \frac{\partial y}{\partial s}\partial_y = e^s\cos t\,\partial_x + e^s\sin t\,\partial_y = x\partial_x + y\partial_y$

$\partial_t = \frac{\partial x}{\partial t}\partial_x + \frac{\partial y}{\partial t}\partial_y = -e^s\sin t\,\partial_x + e^s\cos t\,\partial_y = -y\partial_x + x\partial_y.$

$$\partial_s^2 = \partial_s(x\partial_x + y\partial_y) = (x\partial_x + y\partial_y)(x\partial_x + y\partial_y)$$

$$= x\partial_x(x\partial_x) + x\partial_x(y\partial_y) + y\partial_y(x\partial_x) + y\partial_y(y\partial_y)$$

$$= x(\partial_x + x\partial_x^2) + xy\partial_x\partial_y + yx\partial_y\partial_x + y(\partial_y + y\partial_y^2)$$

$$= x^2\partial_x^2 + y^2\partial_y^2 + 2xy\partial_x\partial_y + x\partial_x + y\partial_y$$

$$\partial_t^2 = (-y\partial_x + x\partial_y)(-y\partial_x + x\partial_y)$$

$$= y\partial_x(y\partial_x) - x\partial_y(y\partial_x) - y\partial_x(x\partial_y) + x\partial_y(x\partial_y)$$

$$= y^2\partial_x^2 - x(\partial_x + y\partial_x\partial_y) - y(\partial_y + x\partial_x\partial_y) + x^2\partial_y^2$$

$$= x^2\partial_y^2 + y^2\partial_x^2 - 2xy\partial_x\partial_y - x\partial_x - y\partial_y$$

$$\Rightarrow \partial_s^2 u + \partial_y^2 u = (x^2+y^2)(\partial_x^2 u + \partial_y^2 u) = (x^2+y^2)\cdot 0 = 0$$

□.

4. 多元Taylor展开的计算.

① $f(x,y) = \ln(1+x+y)$ 在 $(0,0)$ 处展开至 2阶.

用一元 $\ln(1+t)$ 的展开 $\approx t - \frac{t^2}{2} + o(t^2)$

$$f(x,y) = (x+y) - \frac{1}{2}(x+y)^2 + o(x^2+y^2)$$

$$= x + y - \frac{1}{2}x^2 - xy - \frac{1}{2}y^2 + o(x^2+y^2).$$

② $f(x,y) = \sqrt{1-x^2-y^2}$. 在 $(0,0)$ 处展开至 2阶.

$f(x,y) = 1 - \frac{1}{2}(x^2+y^2) + o(x^2+y^2)$ (用 $\sqrt{1+t} \approx 1 + \frac{t}{2} + o(t)$)

③ $f(x,y)=\dfrac{\cos x}{\cos y}$ 在 $(0,0)$ 处展开至 2 阶.

$$\cos x = 1-\frac{x^2}{2}+o(x^2)$$

$$\frac{1}{\cos y} = 1+a\cancel{y}^2+o(\cancel{y}^2)$$ (偶函数, 且 $\frac{1}{\cos 0}=1$).

有 $1=(1+a\cancel{y}^2+o(y^2))\cos y = (1+ay^2+o(y^2))(1-\frac{y^2}{2}+o(y^2))$

对比左右 y^2 系数: $0 = a-\frac{1}{2} \Rightarrow a=\frac{1}{2}$.

$$\Rightarrow \frac{1}{\cos y} = 1+\frac{y^2}{2}+o(y^2)$$

$$\rightsquigarrow f(x,y) = (1-\frac{x^2}{2}+o(x^2))(1+\frac{y^2}{2}+o(y^2))$$
$$= 1-\frac{x^2}{2}+\frac{y^2}{2}+o(x^2+y^2).$$

④ $f(x,y)=\arctan\frac{y}{x}$. $(2,2)$ 处展开至 2 阶.

$$\nabla f = \begin{pmatrix} \frac{-y}{x^2+y^2} \\ \frac{x}{x^2+y^2} \end{pmatrix} \xlongequal{(2,2)} \begin{pmatrix} -\frac{1}{4} \\ \frac{1}{4} \end{pmatrix}.$$

$$\nabla^2 f = \begin{pmatrix} \frac{2xy}{(x^2+y^2)^2} & \frac{y^2-x^2}{(x^2+y^2)^2} \\ \frac{y^2-x^2}{(x^2+y^2)^2} & \frac{-2xy}{(x^2+y^2)} \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & 0 \\ 0 & -\frac{1}{8} \end{pmatrix}.$$

$$\Rightarrow f(x,y) = f(2,2) + (-\tfrac{1}{4},\tfrac{1}{4})\begin{pmatrix} x-2 \\ y-2 \end{pmatrix} + \frac{1}{2}(x-2,y-2)\begin{pmatrix} \frac{1}{8} & 0 \\ 0 & -\frac{1}{8} \end{pmatrix}\begin{pmatrix} x-2 \\ y-2 \end{pmatrix}$$
$$= \frac{\pi}{4} - \frac{1}{4}(x-2) + \frac{1}{4}(y-2) + \frac{1}{16}(x-2)^2 - \frac{1}{16}(y-2)^2 + o((x-2)^2+(y-2)^2)$$

⑤ $f(x,y) = x^{\sqrt{y}}$ 在 $(1,1)$ 处展开至 2阶.

法1. $\nabla f = \begin{pmatrix} \sqrt{y}\, x^{\sqrt{y}-1} \\ x^{\sqrt{y}} \ln x \cdot \frac{1}{2\sqrt{y}} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$\nabla^2 f = \begin{pmatrix} (y-\sqrt{y})\, x^{\sqrt{y}-2} & \frac{\sqrt{y}}{x} \cdot x^{\sqrt{y}} \ln x \frac{1}{2\sqrt{y}} + \frac{1}{2\sqrt{y}} x^{\sqrt{y}-1} \\ \bigcirc & \left(\frac{\ln x}{2\sqrt{y}}\right)^2 x^{\sqrt{y}} + x^{\sqrt{y}} \ln x \cdot \left(-\frac{1}{4y^{3/2}}\right) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}.$$

$$\longrightarrow f(x,y) = f(1,1) + (1,0)\begin{pmatrix} x-1 \\ y-1 \end{pmatrix} + \frac{1}{2}(x-1, y-1)\begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}\begin{pmatrix} x-1 \\ y-1 \end{pmatrix}$$

$$= 1 + (x-1) + \frac{1}{2}(x-1)(y-1) + o\left((x-1)^2 + (y-1)^2\right)$$

法2. $z = \ln f(x,y) = \sqrt{y} \ln x = \sqrt{1+t} \cdot \ln(1+r)$. $\begin{cases} t = y-1 \\ r = x-1 \end{cases} \to (0,0)$

用一元展开. $\sqrt{1+t} = 1 + \frac{t}{2} - \frac{t^2}{8} + o(t^2)$

$\ln(1+r) = r - \frac{r^2}{2} + o(r^2)$

$$\Rightarrow z = \left(1 + \frac{t}{2} - \frac{t^2}{8} + o(t^2)\right)\left(r - \frac{r^2}{2} + o(r^2)\right)$$

$$= r + \frac{tr}{2} - \frac{r^2}{2} + o(t^2 + r^2)$$

再用 $f = e^z = 1 + z + \frac{z^2}{2} + o(z^2)$

$$= 1 + r + \frac{tr}{2} - \frac{r^2}{2} + \frac{1}{2}\left(r + \frac{tr}{2} - \frac{r^2}{2} + o(t^2+r^2)\right)^2 + o(t^2+r^2)$$

$$= 1 + r + \frac{tr}{2} - \frac{r^2}{2} + \frac{r^2}{2} + o(t^2+r^2)$$

$$= 1 + r + \frac{tr}{2} + o(t^2+r^2)$$

$$= 1 + (x-1) + \frac{1}{2}(x-1)(y-1) + o\left((x-1)^2 + (y-1)^2\right).$$